

CE 402: Geotechnical Engineering II

Course Code	CE 402
Course Name	Geotechnical Engineering II
L-T-P-C	2-1-0-3
Prerequisites	CE302: Geotechnical Engineering – I or equivalent
Intended for	B. Tech 3rd or 4th Year
Distribution	Discipline Elective
Approval	19th BoA (Nov 2017); 15th Senate (Feb 2018)

Course Contents

- **Subsoil Exploration**
Methods of subsoil exploration Direct, semi direct and indirect methods, Soundings by Standard, Dynamic cone and static cone penetration tests, Types of boring, Types of sample; Criteria for undistributed samples, Transport and preservation of sample, planning of exploration programmes, report writing. [8 Lectures]
- **Earth Pressure**
Types of Earth pressure, Rankines and Coulumb's Active and Passive earth pressure in soils, concepts for slope stability and soil retaining wall design. [6 Lectures]
- **Bearing Capacity**
Safe bearing capacity and allowable bearing pressure, General and local shear failure conditions, Terzaghi's bearing capacity equations its modifications for square, rectangular and circular foundation, Factors affecting bearing capacity of soil. [6 Lectures]
- **Design of Shallow Foundations**
Factors effecting location of foundation and design considerations of shallow foundations, choice of type of foundations, Foundations of expansive soils. [6 Lectures]
- **Settlement Analysis**
Causes of settlement, computation of settlement, immediate and consolidation settlement, allowable settlements, Measures to reduce settlement. [4 Lectures]
- **Pile Foundations**
Types, Construction, load carrying capacity of single pile – Dynamic formula, Static formula, Pile load tests, Load carrying capacity of pile groups, settlement of pile groups, Negative skin friction. [12 Lectures]

Text Books

1. Braja M. Das, **Principles of Foundation Engineering**, Cengage Learning, 2015.
2. V. N. S. Murty, **Geotechnical Engineering**, Book World Enterprises, 2008.

References

1. W. C. Teng, **Foundation Design**, Printice Hall Publishers, 1962.
2. J. E. Bowles, **Foundation Analysis and Design**, Tata McGrawhill Publishers, 1997.
3. Shamsheer Prakash, Gopal Ranjan and Swami Saran, **Analysis, Design of Foundations and Earth retaining structures**, IBH Publishers, 1979.